



CANDIDATE
NAME

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CLASS

2T

INDEX
NUMBER

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BIOLOGY

Paper 2 Core Paper

9747/02

Wednesday 2 September 2009
2 hours

Additional materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and HT group on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams, graph or rough working.
Do not use paper clips, highlighters, glue or correction fluid.

Section A Structured Questions

Answer **all** questions.
Write your answers in the spaces provided on the question paper.
Show All working

Section B Essay Questions

Answer one question only.
Write your answers on the separate answer paper.
All working for numerical answers must be shown.

At the end of the examination,

1. fasten all your work securely together;
2. enter the number of the Section C question you have answered in the grid opposite.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	80
1 [11]	
2 [12]	
3 [12]	
4 [12]	
5 [9]	
6 [15]	
7 [9]	
Section B	20

Section A

Answer all questions in this section

- Brazil is one of the world's largest producers of tropical fruits such as oranges, pineapples, bananas, acerolas and persimmons amongst others. Particularly in the Northeast region, cashews are cultivated on a large scale. However, due to the higher commercial value of the cashew nut, only 15% of the juicy portion (*peduncle*) is utilized for the derived products. This is an unbearably great loss for the region, which is poor in food resources. Fermentation, however, can be a viable solution to avoid this loss. This includes alcoholic and acetic, as well as various other types of fermentation. *Sacharomyces cerevisiea* bakers yeast was used for cashew wine fermentation. Fig 1.1 shows the Fermentation kinetics of Cashew wine production

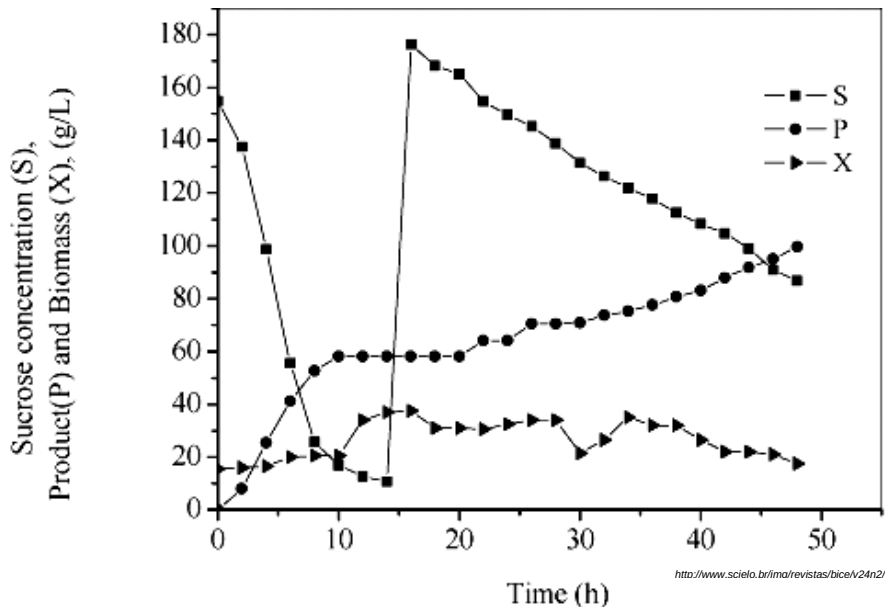


Fig 1.1: Fermentation kinetics of Cashew wine production: concentration of substrate (S), concentration of ethanol (P) and concentration of biomass (X) (expressed in g/L).

- (a) Explain the “fermentation” (Line 5 of introduction paragraph) and outline the process and conditions by which this takes place. [5]

1. Fermentation is the process of **Respiration** under **anaerobic conditions** (absence of O_2)
2. **Glucose is the main substrate** and under **Glycolysis** produces **pyruvate** at the expense of **ATP** and **NADH**
3. **Pyruvate** is then **converted to ethanol** under the action of alcohol dehydrogenase at the expense of **NADH**
4. the **regeneration of NAD^+** allows for Glycolysis to continue
5. **fermentation** yields **2 ATP** as a result.

Marking Comments:

1. Some mistakes made in confusion between NADH and NADPH.
2. There is no need to mention Lactic acid fermentation. Since only Alcoholic fermentation only takes place in Bacteria.
3. Qn asked to EXPLAIN, and outline the Process- most did not.

(b) With reference to Fig 1.1, explain fully the production of ethanol over time. [6]

1. From **0-10hrs** there is seen an **exponential increase** in ethanol concentration from **0-60g/L** of a **magnitude of 60g/L**.
2. exponential rise due to the **exponential increase in population** of *Sacharomyces cerevisiea* bakers yeast from **18 to 20g/L** from **0 to 10hrs**
3. from **10-15hrs** sees **no increase** in ethanol concentration at **60g/L**.
4. corresponding to a **low concentration of substrate 20-18g/L** due to the further increase in population of yeast during the same time frame, **competition for resources is high**.
5. from **15-50hrs** sees a **linear increase** in ethanol production from **60-100g/L** of **40g/L**.
6. corresponding to an average production of ethanol as **population of yeast stabilizes** at an **average of 20g/L**.

Marking Comments:

1. Most didn't quote data from graph.
2. there needed to be an EXPLANATION for the rise and fall of the graph.
3. Perspective is BEST seen from the PRODUCTION OF ETHANOL rather than substrate.
4. Poor phrasing resulted in few marks.

[Total : 11]

2 Fig 2.1 shows a section of DNA *undergoing replication*.

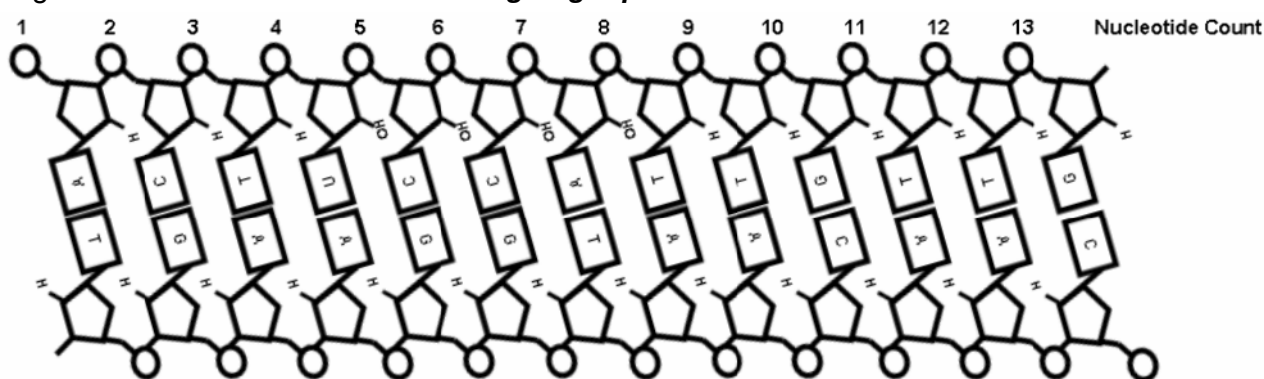


Fig 2.1 Section of DNA during Replication

(a) Which strand of the DNA molecule is being synthesized? Give reasons for your answer. [3]

1. Strand being synthesized: Lagging strand
2. reason: Presence of **only deoxyribose nucleotides** in **lower strand ending in 5'** indicating parent template
3. reason: Presence of **ribose nucleotides** in between **deoxyribose nucleotides** in **upper strand ending in 3'** indicating Okazaki fragment

Marking Comments:

1. reject antisense and sense since neither are being synthesized and are not called that even upon completion.

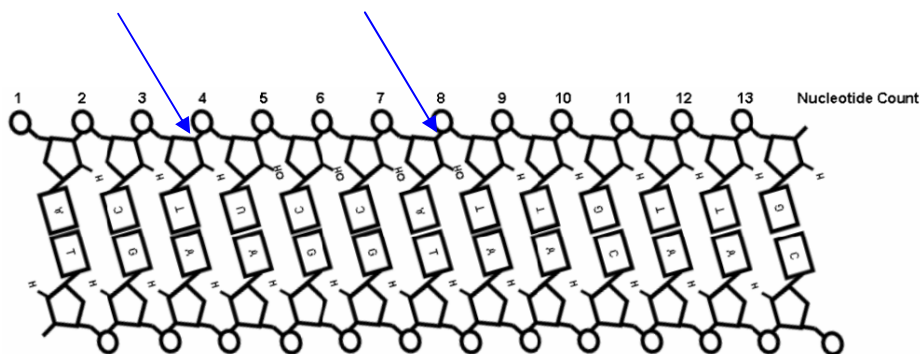
(b) Name and describe fully, the function and mechanism of the enzyme responsible for further processing of nucleic acid strands shown in Fig 2.1. [6]

1. Enzyme: **DNA Polymerase I**
2. Function #1: **Proof reading** which entails the **removal of ribose nucleotide Primers** and **replacement** with corresponding **deoxyribose nucleotides**.
3. Function #2: **DNA Repair** also includes the **removal of mispairings** that may occur as a **result of mistakes in function of DNA Polymerase III**, removal of these **deoxyribose nucleotides** and replacing with **complementary deoxyribose nucleotides**.
4. Using **5' Parent strand** as a **Template** and replaces the **excised nucleotides via Watson and Crick complementary base pairing**.
5. Requirement of a **free 3'OH group** provided by **okazaki strand** preceeding gap
6. formation of **phosphodiester bond** thru **condensation** and **energy** derived from **hydrolysis of nucleoside triphosphates**.

Marking Comments:

1. enzyme cannot be DNA Poly III due to phrase: "Further Processing" in Qn.

(c) Indicate with **two arrows**, on Fig 2.1 where phosphodiester bonds will be broken as a result of the enzyme mentioned in 2(b). [1]

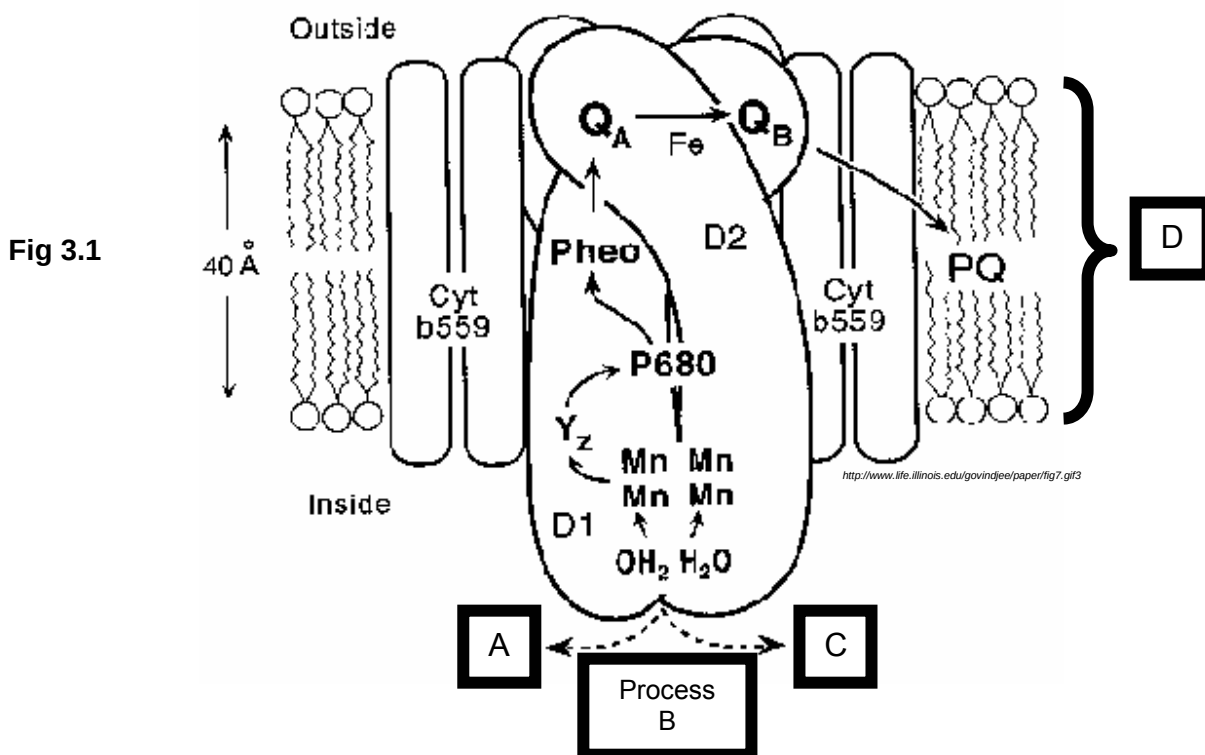


(d) Give 1 difference in the action of RNA Polymerase II and the enzyme mentioned in 2(b). [2]

1. **RNA Polymerase II** works only on **one template strand of DNA** at any one time in Transcription.
2. **DNA Polymerase I** works on **both template strand of DNA** in the **removal of primers**.

[Total: 12]

3 Fig. 3.1 shows a single photosystem within a chloroplast.



(a) With reference to Fig 3.1, complete in detail, the table below: [3]

	Identify	Function or use	Mk
A	1] Oxygen (O₂)	1] Resulting by-product , may be used in respiration	1
B	2] Photolysis of Water	2] water is split to derive free electrons to replenish / reduce PSII	1
C	3] Hydrogen ions (H⁺)	3] used in the production of NADPH₂	1

(b) With reference to Fig 3.1, explain the photoactivation of chlorophyll. [2]

1. Photoactivation of chlorophyll is the process by which **Light energy excites the electrons** within Photosystems
2. till these **electrons raised to a higher energy state** and are **promoted**, to be accepted by primary electron acceptors, leaving chloroplasts in an oxidized state.

(b) Name and describe the function of structure **D** in photophosphorylation. [2]

1. **D: Thylakoid Membrane**
2. **acts as a reservoir for H⁺ ion build up for Chemiosmosis in the synthesis of ATP.**

(c) List two ways in which cyclic photophosphorylation differs from non-cyclic photophosphorylation. [3]

1. **in cyclic photophosphorylation energized electron originates from the promotion from PS700 whereas in non-cyclic photophosphorylation electron originates from from the promotion from PS680.**
2. **During cyclic photophosphorylation, electrons from photosystem - I are not passed to NADP from the electron acceptor. Instead the electrons are transferred back to P700 therefore NADPH₂ are not formed during cyclic photophosphorylation.**
3. **Since only PS700 is only used in cyclic photophosphorylation therefore PS680 is not involved and photolysis of water does not occur therefore oxygen also is not formed during cyclic photophosphorylation.**

(d) Describe fully how the products of cyclic photophosphorylation are used and explain the function of cyclic photophosphorylation. [2]

1. **ATP is used in the Calvin cycle**, specifically in the conversion of **Glycerate 3 Phosphate** (reject GP) to **Glyceraldehyde 3 Phosphate** or **Triose Phosphate** (reject GALP or TP)
2. **ability to synthesize greater amounts of ATP to compensate for the demands of the light independent reaction** or Calvin cycle.

[Total: 12]

Marking Comments:

1. **no point discussing NADPH since NADPH is not a product of Cyclic Photophosphorylation .**

- 4 Pingelap is an atoll in the Pacific Ocean, part of Pohnpei state of the Federated States of Micronesia, consisting of three islands: Pingelap Island, Sukoru and Daekae, linked by a reef system and surrounding a central lagoon, although only Pingelap Island is inhabited, an island in Micronesia.

[**Micronesia** , is a subregion of Oceania, comprising hundreds of small islands in the Pacific Ocean]

In approximately 1775, a typhoon reduced the population of the island to only 20. Among survivors, one of them was heterozygous for achromatopsia.

[**Achromatopsia** , is a medical syndrome that exhibits symptoms relating to at least five separate individual diseases. Although the term may refer to acquired disorders such as color agnosia and cerebral achromatopsia, it typically refers to an autosomal recessive congenital color vision disorder, the inability to perceive color and to achieve]

After few generations, the prevalence of achromatopsia is 5% of population and 30% as carriers (by comparison, in the United States, only 0.003% of population have complete achromatopsia).

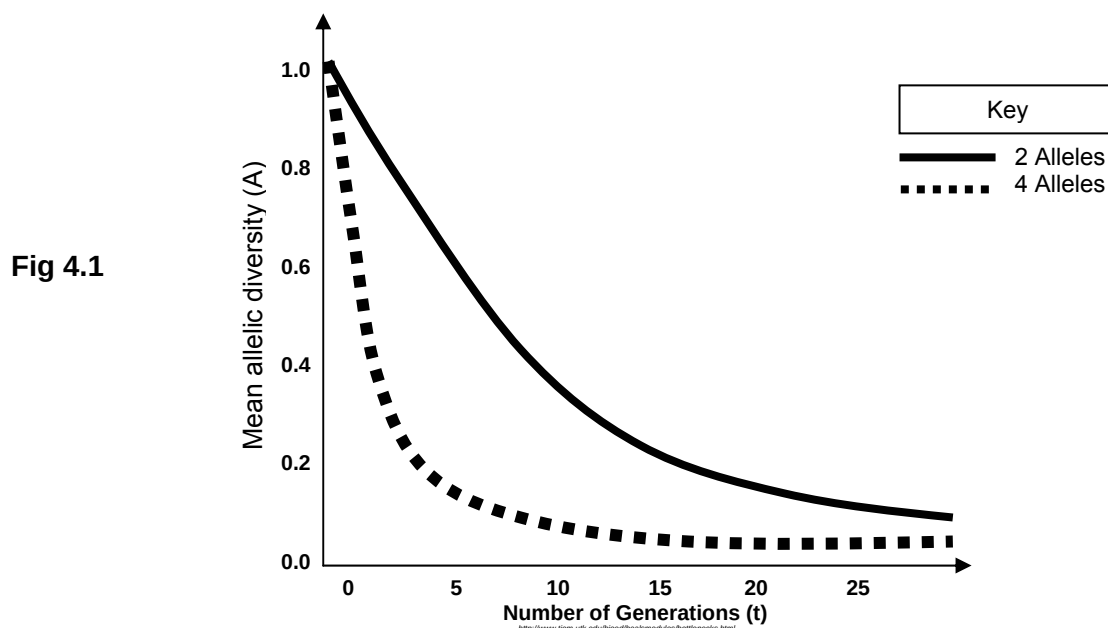
(a) Explain in detail, from an evolutionary stand point what has happened to the inhabitants of Micronesia. [5]

1. The **Bottle Neck effect (accept Genetic Drift)** has taken place, with the **main population severely reduced** in size due to a **single event** may leave only a **small fraction of population to survive and reproduce**.
2. that **causes diversity / Gene Pool to also be diminished** due to the **permanent elimination of alleles**.
3. **Geographical isolation** prevents gene flow into population.
4. **frequency of recessive alleles increase** as seen from **5% to 30%**
5. as a result of **inbreeding**.

Studies have been made on the effect of allelic diversity in populations experiencing the same conditions and circumstances as Micronesia, over time.

(**Allelic diversity**- measure of genetic diversity based on the average number of alleles per locus present in a population).

Fig 4.1 shows the allelic diversity across subsequent generations after a cataclysmic event.(e.g. disaster) Populations of 2 and 4 alleles are assumed to be the same size.



(b) with reference to Fig 4.1, explain fully the **general trend** observed in the population with 2 alleles over time. [5]

1. **Decline in 2 Allele population from 1.0A to 0.6A over 5 generations is steep**
2. **decline is greater** at the beginning since in a bottleneck event, **alleles are removed permanently from the gene pool** and therefore **allele diversity drops drastically**.
3. After the bottleneck event, **inbreeding occurs** and allele frequency changes, **homozygous recessive states become more frequent**
4. but **homozygous recessive states are less advantageous and die off**, general trend therefore sees a **decline in allelic diversity** over generations.
5. but this **decline is less steep** due to the **population also growing in numbers** and **allele frequency rises** so that even if homozygous recessive traits are less advantages, allele diversity does not change drastically.

(c) With reference to Fig 4.1 explain the difference between trends in populations with 4 alleles from populations with 2 alleles. [2]

1. The **greater no. of alleles, the more variation** but **allele frequency is lower**, therefore is more severely hit after the event, showing a **greater decline in variation due to the elimination of alleles**.
2. with **2 alleles**, and with populations of the same size, **allele frequency is higher**. Therefore after event loss even with elimination of individuals, gene representation is not as affected, showing a **gradual decrease in variation**.

2 Alleles: HW Eqn: $P^2 + 2PQ + Q^2$	
Allele Freq	P: $4/8 = 0.5$
	Q: $4/8 = 0.5$
3 Alleles: HW Eqn: $P^2 + Q^2 + R^2 + 2PQ + 2QR + 2PR$	
Allele Freq	P: $6/18 = 0.3$
	Q: $6/18 = 0.3$
	R: $6/18 = 0.3$
With Popln same size Freq will be even smaller for greater allele Diversity	

Bottleneck

[Total:12]

- 5 In *Drosophila* flies, a gene for wing shape found on chromosome 2 has two alleles. Curly wings were found to be dominant over normal wings. Like in Humans, gender in *Drosophila* have Female flies as homogametic while males are heterogametic. The gene for *Drosophila* eye colour is sex linked and has two alleles – the allele for red eyes being dominant to the allele for white eyes.

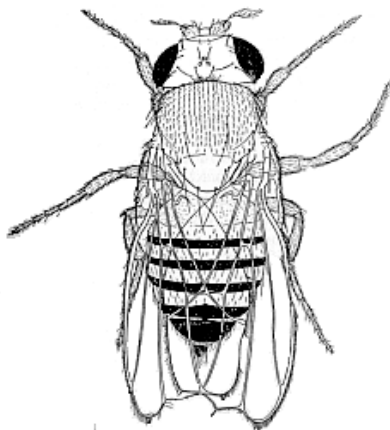


Fig 5.1: Normal Wings

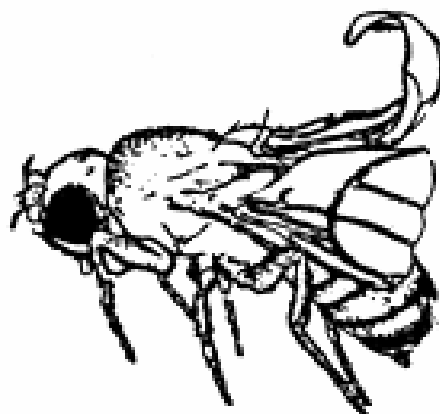


Fig 5.2 Curled Wings

- (a) Draw a genetic diagram showing the phenotypes of the offspring and the expected ratio of phenotypes when a white eyed female is crossed with a red eyed male, both are heterozygous for curly wings. [4]

Defining Alleles

A: Curly Wings

a: Normal Wings

R: red Eyes

r: White Eyes

Gender	Female		Male
Parental genotype:	$X^rX^r Aa$	x	X^RYAa
Meiosis			
Gametes:	X^rA X^ra		X^RA X^Ra YA Ya

	X^RA	X^Ra	YA	Ya
X^rA	X^RX^rAA	X^RX^rAa	X^rYAA	X^rYAa
X^ra	X^RX^rAa	X^RX^raa	X^rYAa	X^rYaa

All correct [1]

Offspring Genotype	X^RX^rAA 2(X^RX^rAa)	X^RX^raa	X^rYAA 2(X^rYAa)	X^rYaa
Offspring Phenotype	Red eyed female with curly wing	Red eyed female with normal wing	White eyed male with curly wing	White eyed female with normal wing
Phenotype Ratio	3	1	3	1

(b) 240 offspring were obtained from the above cross, and the phenotypes of the offspring were as follows:

- 83 red-eyed females with curly wings
- 41 red-eyed females with normal wings
- 78 white-eyed males with curly wings
- 38 white-eyed males with normal wings

Degrees of freedom	Probability, p				
	0.10	0.05	0.02	0.01	0.001
1	2.71	3.84	5.41	6.64	10.83
2	4.61	5.99	7.82	9.21	13.82
3	6.25	7.82	9.84	11.35	16.27
4	7.78	9.49	11.67	13.28	18.47

χ^2 test: $\chi^2 = \sum \frac{(O - E)^2}{E}$ $v = c - 1$

Key to symbols: Σ = sum of ...
 v = degrees of freedom
 c = number of classes
 O = observed value
 E = expected value

Use a Chi-square test to determine if the above cross follows mendelian ratios and explain your results.

	red-eyed females with	red-eyed females with normal	white-eyed males with curly	white-eyed males with normal wings
--	-----------------------	------------------------------	-----------------------------	------------------------------------

	curly wings	wings	wings	
Observed	83	41	78	38
Expected	90	30	90	30
O-E	7	11	12	8

$$\begin{aligned}
 X^2_{cal} &= 7^2/90 + 11^2/30 + 12^2/90 + 8^2/30 \\
 &= 0.54 + 4.03 + 1.60 + 2.13 \\
 &= 8.31
 \end{aligned}$$

(Calculations / Workings [1] Correct Answer [1])
[1]

$$X^2_{critical} = 7.82$$

Norm: $X^2_{cal} < X^2_{critical}$

Since $X^2_{cal} > X^2_{critical}$ 8.31 is greater than 7.82 for 95% Probability (P)

or

Probability of getting Chi² value of 8.31 is between p=0.05 and p=0.02, P<0.05 (or less than 5% probability)

(either explanation [1])

Conclusion

Cross **does not** follow **Mendelian ratios**

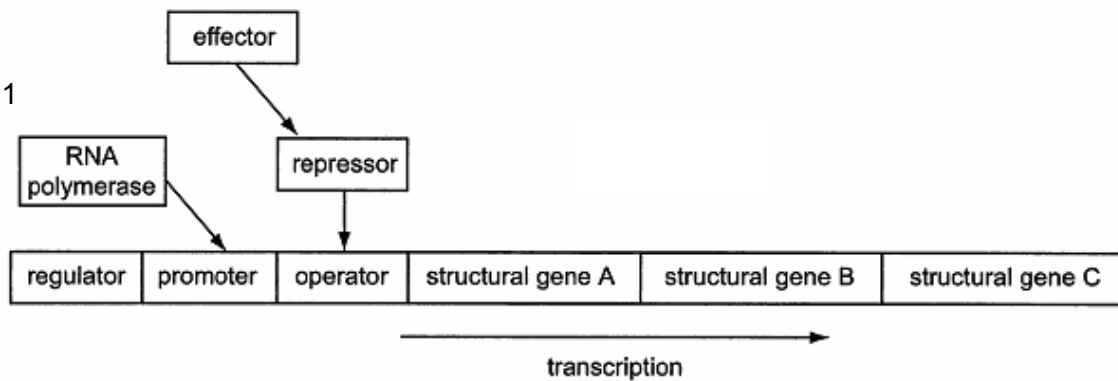
Reason: may be due to **epistasis** or other external factors [Reject Linkage]

(either explanation [1])

[Total 9 marks]

6 Fig 6.1 shows some of the molecules involved in controlling transcription in bacteria.

Fig 6.1



The polypeptide synthesized from the structural gene A acts as an effector. The effector binds the repressor to the operator to block transcription.

(a) (i) Which molecule acts as an end product inhibitor and why? [1]

1. **Effector**, as it **binds the repressor to the operator** to block transcription.

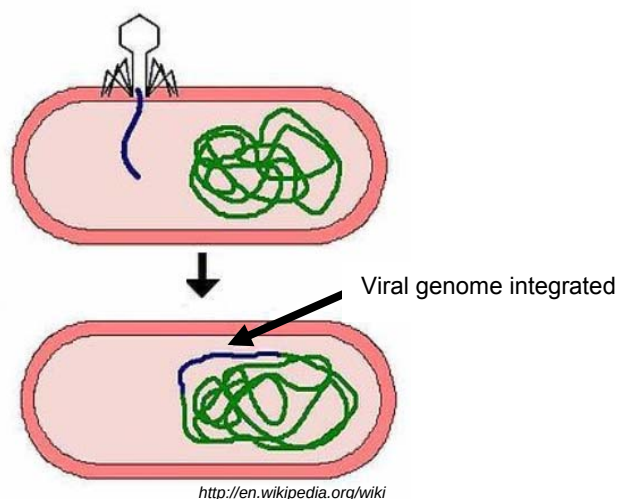
(ii) Explain, using a **named example**, how an operon as shown in Fig 6.2 can be used to control transcription. [4]

1. In ***trp* operon** (*trp*: tryptophan) used for regulation of **synthesis of repressible enzymes** which catalyze an anabolic pathway for producing tryptophan.
2. Transcription of the operon occurs because the **repressor is unable to bind to the operator region**.
3. After a **corepressor** (*trp*) binds to the repressor, the **repressor now binds to the operator and blocks transcription**.
4. mRNA and the proteins it codes for are **not** made.

(b) (i) Distinguish between structural and regulatory genes in relation to a **named operon**. [3]

1. A structural gene is a region of DNA that **codes for a protein or RNA molecule** that forms part of a **structure** or has an **enzymatic function**
2. e.g. *lacY*, *lacZ*, *lacA*, (Transacetylase / Permease / Beta –Galactosidase) but excludes *lacI*.
3. A regulatory gene codes for a specific **protein product** that **regulates the expression** of the **structural genes** (e.g. *lacI*).

Fig 6.2 shows a virus attacking a cell.



- (c) (i) Based on Fig 6.2, identify the type of virus shown and provide a **named** example. [1]
 1. **Bacteriophage, e.g. lambda phage.**

- (ii) With reference to Fig 6.2, describe the reproductive cycle of this virus using the **named** example in (c) (i). [6]

1. Phage λ has **one short tail fiber** (Spike) which attaches to a host cell and **inject its DNA**.
 2. Phage DNA **circularizes** and certain factors determine whether **lytic or lysogenic cycle is entered**.
 3. phage DNA **integrates** into the bacterial chromosome, becoming a **prophage** as in Fig 6.2
 4. The bacterium **reproduces normally**, with **prophage genome integrated** and transmitting it to the daughter cells.
 5. Occasionally, a prophage exits the bacterial chromosome, initiating a **lytic** cycle.
 6. in which the host mechanisms are taken over in the synthesis of Phage λ copies.
- (must include point 3 and any other 4 points in sequence)

[Total: 15]

7 Fig 7.1 shows a section of DNA, containing one gene.

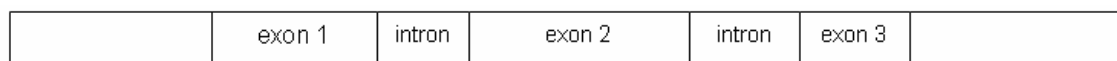


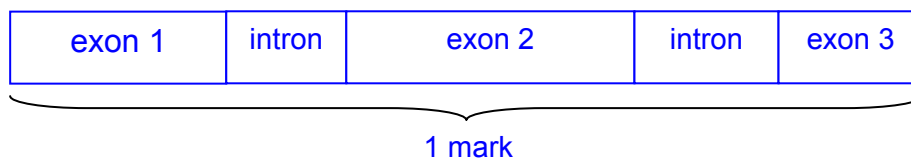
Fig 7.1

- (a) (i) State one way in which the structure of this gene indicates that it is from a eukaryote. [1]

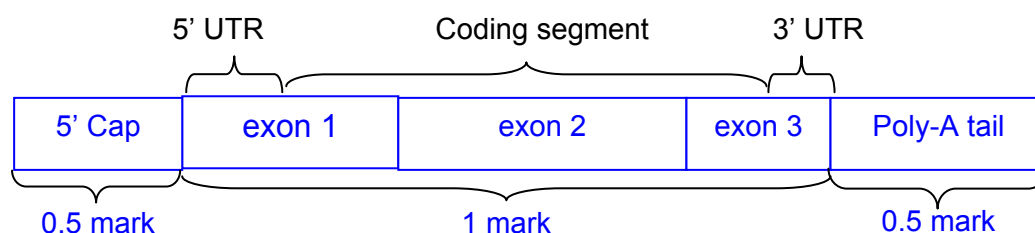
1. **Presence of introns, non-coding DNA sequences.**

- (ii) Using the same diagrammatic format as Fig 7.1, show what functional elements of Fig 7.1 will be mapped out in the pre-mRNA and mRNA. [3]

pre-mRNA



mRNA



- (b) Describe three post-transcriptional modifications to pre-mRNA. [5]

1. In the eukaryotic nucleus, the introns are excised and the exons are spliced together through a process known as **RNA splicing**.
2. Presence of **splice site**, at each end of an intron which forms the signal for RNA processing and are recognized by **small nuclear ribonucleoproteins (snRNPs)**.
3. snRNPs and other proteins form a molecular complex called **spliceosome**. which **interacts** with the splice sites at the ends of an intron and **cuts** at specific points to **release the intron** and **joins** together the **two exons that flanked the intron**.

4. An example of regulation of gene expression at RNA splicing is **alternative RNA splicing**, where different mRNA molecules are produced from the same primary transcript, depending on which RNA segments are treated as **exons** and which as **introns**.
5. In each cell type, **specific regulation proteins** would control intron-exon choices by binding to regulatory sequences within the primary transcript.

(max 3 marks)

6. The 5' end of a pre-mRNA is synthesized first and it is capped with a modified form of a **guanine (G) nucleotide** onto the 5' end **after** transcription of the first 20 to 40 nucleotides.
7. Before the mRNA exits the nucleus, the 3' end of a pre-mRNA is added with a **poly-A tail** consisting of 50 – 250 **adenine nucleotides**.

[Total: 9]

Section B

Answer **only one question** in the following section

Write your answers on separate answer paper provided.

Your answer should be illustrated by large, clearly labeled diagrams, where appropriate.

Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections (a), (b) etc., as indicated in the question.

- 8 (a) Compare and contrast Neo-Darwinism with Neutral Theory [4]

Similarities

1. Both Neo-Darwinism (ND) and Neutral Theory (NT) show that Advantages mutations are favored and selected while disadvantages traits are removed during selection.

Differences

Frequency of appearance of genes,

2. **ND in Natural selection** (Selection theory) depends **more on selection** whereas
3. in NT frequency of alleles is dependent on **genetic drift**.

Evolution

4. Evolution of population due to **selection of advantageous alleles** as in ND, however
5. in NT evolution is due to mainly **neutral alleles**

Mutations

6. Mutations in ND are defined as **Good/Bad mutations**, however
7. in Neutral theory, mutations are categorized as **Good, Neutral and Bad**

- (b) Explain how biogeography and the fossil record support the evolutionary deductions based on homologies. [6]

1. **Evolution** involves the **study of descent with modification**, a comparison between current organisms and those of the past in order to draw conclusions about the relationships between extinct species and modern day species. The history of life on earth can be documented via **fossil records** and **biogeography**.

Fossil record

2. Fossils also allow evolutionary biologists to study the structure of **extinct species** in order to discover possible **evolutionary relationships** between past and present species based on **homology**.
3. e.g. **Archaeopteryx lithographica**- comparative study of similarities between modern day birds and reptiles.
4. e.g. **Hominid Evolution**- comparative study of modern day humans and their fossil lineage.

Biogeography

5. Biogeography the study of the **geographical distributions of animals and plants, including fossils**. These studies sometimes provide evidence for evolution and natural selection
6. e.g. **Wallace's Line**. To the west of Wallace's Line (Borneo and the Malayan Peninsula) were birds with Asian features and to the east (New Guinea and Australia) were birds with distinctly Australian features.
7. e.g. **The Continent of Pangaea** Pangaea was theorised to be the **proto-continent** that existed before the movement of the tectonic plates separated it into the continents we know today, The separation of the continents would explain why some species on different continents share common phylogenetic lineages. It also provides the necessary **geographical isolation** that results in speciation via evolution.
8. Evidence for this include: **Fossils of reptiles** found in **South Africa** were also found in **Brazil** and **Argentina**. Fossils of amphibians and reptiles found in **Antarctica** were also found in **South Africa, India, and China**. Most of the marsupial mammals alive today are confined to **South America** and **Australia**

8 (c) Compare and contrast transcription and translation

[10]

Similarity

1. Both processes occur within the **cytoplasm** of the cell.
2. Both processes involve **Watson and Crick complementary base pairing** as part of synthesis.

Differences

On the level of template

3. Translation utilizes an **mRNA as a template**
4. as opposed to **DNA as a template** in Transcription

On the level of energy

5. Translation utilizes **ATP in the synthesis of activated amino acids** (reject activated tRNAs) and
6. **GTP in the action of peptidyl transferase and hydrolysis involving the release factor**
7. Transcription utilizes **energy derived from hydrolysis of ribonucleoside triphosphates**

On the level of Bond formation

8. Translation results in **peptide bonds** being formed between subunits
9. as opposed to **phosphodiester** bonds formed in transcription.

On the level of Enzymes

10. Enzymes used in translation are **Amino acyl-tRNA synthetase** and **peptidyl-transferase**
11. Enzymes used in transcription are **RNA polymerase II**

On the level of product

12. Translation produces a **primary polypeptide**
13. as opposed to transcription which produces an **mRNA**

On the level of raw materials

14. Translation utilizes **activated amino acids**
15. as opposed to transcription which utilizes **ribonucleoside triphosphates**

[any 10]

[Total: 20]

On level of the overall **Structure of Genome**

1. Prokaryotes have **no nuclear membrane**, genome exist as a **nucleoid** / simultaneous transcription and translation can occur / fewer levels of control of gene expression.
Eukaryotes have **nuclear membrane**, genome is enclosed within it / Absence of simultaneous transcription and translation / more levels of control of gene expression.
2. Prokaryotes have **only one origin of replication** as they have a smaller genome while eukaryotes have **multiple origins of replication** due to more chromosomes present.

On the level of **Size**

3. Prokaryotes: Smaller genome and smaller number of bases / smaller number of genes / coding regions.
Eukaryotes: More than one chromosome and larger number of chromosomes and bases / genome / larger number of genes / coding regions.
4. Prokaryotes: Single chromosome, usually haploid (with only 1 set of chromosome) / no pairing of homologous chromosomes possible.
Eukaryotes: Multiple chromosomes, usually diploid (with 2 sets of chromosomes) / pairing of homologous chromosomes possible / Enables genetic recombination during meiosis.

On the level of **Packing of DNA**

5. **Prokaryotic DNA is naked** while eukaryotic DNA is arranged in **nucleosomes wrapped around histones**.
6. Prokaryotic genome is initially packed to form loop domains and undergoes further DNA supercoiling to form the nucleoid.
Eukaryotes have an **elaborate, multilevel system** of DNA packing to fit all the DNA into the nucleus in preparation for cell division / 10 nm fiber to 30 nm chromatin fiber or solenoid to looped domain forming 300 nm fiber to metaphase chromosome.
7. The compacting of DNA in prokaryotes is with the aid of RNA and nucleoid proteins and these proteins differ from the histone proteins present in eukaryotic nuclei. / Does not achieve same level of condensation complexity as eukaryotes.
DNA packaging in eukaryotes is with the aid of histone proteins and non-histone scaffolding proteins. / Allows for increased structural complexity, folding to higher degree of condensation. / Rate of transcription is controlled by allowing for increased DNA condensation / conversion between euchromatin and heterochromatin states.

On the level of **Linearity and Circularity**

8. DNA in prokaryotes occurs in a **loop**, i.e. circular while eukaryotic DNA is **linear**.

On the level of **Presence of Introns / Telomeres / Centromeres**

9. Prokaryotes **do not have introns**, so there is **no splicing**, hence only one possible protein product per gene.
Eukaryotes **have introns** which allows for **alternative splicing** / Different mRNA molecules are produced from the same primary transcript.
10. Prokaryotes **do not have telomeres** due to their **circular DNA** and they do not have centromeres as well.
Eukaryotes have **telomeres protect** the ends of chromosomes from **degradation and fusion with other chromosomes** and they have centromeres for kinetochore formation to anchor spindle fibres so that they could be pulled to the poles.

On the level of **Type of Regulatory Sequences**

11. Prokaryotes **do not have control elements** while eukaryotes **have control elements** which are grouped into either **silencers** or **enhancers** to enable **better control of gene expression** as transcription of a particular gene can be upregulated or downregulated.

On the level of Mechanism used to Control Transcription

12. Basic mechanism of controlling gene expression in bacteria is known as the **operon model** where an **operon** is a **cluster of genes whose expression is controlled by a single operator**.

Eukaryotes **have control elements** which are grouped into either **silencers** or **enhancers** to enable **better control of gene expression** as transcription of a particular gene can be upregulated or downregulated. / Each eukaryote gene is transcribed separately, with separate transcriptional controls on each gene.

13. Prokaryotes have **polycistronic genes** / allows coordinated control of metabolic enzymes involved in the same pathway / all products of operon are produced in the same amount.
Eukaryotes have **monocistronic genes** where individual enzymes are controlled individually by separate promoters / different enzymes can be produced in different amounts or not produced at all.

(Any 10 points, must have comparison and comparison must be done at the same level otherwise no marks awarded)

- 9 (b) Define control elements and explain how they influence transcription. [10]

1. **Control elements** are **noncoding DNA segments** that regulate transcription of a gene by binding to proteins like **transcription factors**.
2. Control of gene activity in eukaryotic cells usually involves **a balance between the actions of transcriptional activators and repressors**.
3. Binding of **transcriptional activators** to specific **DNA regulatory sequences called enhancers** turns on transcription.
4. Binding of **transcriptional repressors** to other **DNA regulatory sequences called silencers** turns off transcription.
5. **Enhancers** are a segment of **eukaryotic DNA** containing **multiple control elements**, usually located **far** from the gene whose transcription it regulates.
6. Rate of gene expression is strongly increased by the binding of proteins - **activators** such as **specific transcription factors** to the control elements of enhancers.
7. Activator proteins known as specific transcription factors bind to distal control elements grouped as an enhancer in the DNA.
8. **Protein mediated bending of DNA** bring the **bound activators in contact with mediator proteins** which in turn interact with **general transcription factors at the promoter**.
9. These **multiple protein-protein** interactions help assemble and position the **transcription initiation complex** on the promoter.
10. RNA synthesis is **initiated** and rate of transcription of the gene is **increased**.

[Total: 20]